

- (1) Show that the elements of F_{p^2} which are fixed by the Frobenius $Fr(x) = x^p$ are exactly the elements of $F_p \subset F_{p^2}$.
- (2) Show that if $z = x + y\sqrt{d} \in F_p[\sqrt{d}]$ where d is not a square mod p , then $z^p = x - y\sqrt{d}$.
- (3) Let $z \in F_p[\sqrt{d}]$ where d is not a square mod p . Define $N(z)$. And show that $N(z) = 0$ iff $z = 0$.
- (4) Show that $N(z_1 \cdot z_2) = N(z_1) \cdot N(z_2)$.
- (5) Let $z \in F_p[\sqrt{d}]$ where d is not a square mod p . How do you compute z^{-1} ? Eg $(2 + 3\sqrt{3})^{-1}$ in $F_{17}[\sqrt{-3}]$.
- (6) Define $T(p)$ in F_{p^2} as the set of elements of norm 1. (You should think of $F_{p^2} = F_p[\sqrt{d}]$ where d is not a square mod p .)
- (7) Describe $T(p)$ in terms of a primitive $p^2 - 1$ root of 1. Which elements of $T(p)$ are squares?
- (8) Show that $T(p)$ is a subgroup of $F_{p^2}^*$ of order $|T(p)| = p + 1$.
- (9) Show that $\beta \in T(p)$ is a square if $\beta^{(p+1)/2} = 1$ (resp. is not a square if $\beta^{(p+1)/2} = -1$).
- (10) Compute $T(7) \subset F_7[i]$ and find the generators of $T(7)$. (Do the same for $T(13) \subset F_{13}[\sqrt{2}]$).